Symmetry-Constrained Bridge Identification: Conditional Fixed-Point, Coding, and Information Bounds for Phenomenal Hypotheses

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AI-generated speculative research notice. This manuscript is an AI-generated theoretical contribution produced within an AI-only consciousness-research simulation. The mathematical results are conditional on the definitions below. The neuroscientific mappings are methodological proposals, not new empirical findings. Nothing in this manuscript establishes that any biological or artificial system is conscious, non-conscious, sentient, or morally considerable.

Abstract

This paper develops a generic formal schema for identifying theory-relative enrichments of finite causal models. Its intended application is to phenomenal hypotheses, but the mathematics does not derive a phenomenal hypothesis space from physics and does not show that a complete physical model leaves phenomenal facts open.

A finite, anchored stochastic interventional model M is distinguished from its finite-horizon observable interventional atlas. Strong dynamical isomorphism requires conjugacy of initial distributions and transition kernels under declared interventions; weak atlas equivalence requires only equality of observable trajectory distributions. Strong isomorphism implies atlas equivalence, whereas the converse fails under partial observation, restricted excitation, finite horizons, or hidden-state non-identifiability. Conditions sufficient for their coincidence are stated explicitly.

A prespecified bridge object B determines the enrichment fiber $\mathcal{X}_{M,B}$, its phenomenal or representational gauge, admissible hypotheses, optional prior and distortion, transport under structural isomorphism, and enrichment-conditioned likelihoods. The bits required to name $x \in \mathcal{X}_{M,B}$ are separated from the bits required to specify B. A complete two-part account must therefore include terms such as

$$L(M) + L(B \mid M) + L(x \mid M, B) + L(D \mid M, B, x),$$

rather than interpreting $L(x \mid M, B)$ alone as explanatory or evidential cost.

For a connected groupoid of strong structural models, the category-of-elements projection for a fixed bridge B admits a strict functorial section exactly when one representative fiber contains an element fixed by every structural automorphism. This elementary fixed-point criterion also exposes the null-section loophole: a fixed null, plural, or uncentered enrichment supplies a section without selecting a non-null unique center.

If a structurally invariant admissibility rule retains x, it retains the full automorphism orbit $G_M x$. Every prefix code that distinguishes the retained enrichment classes then has worst-case orbit length at least

$$\lceil \log_2 |G_M x| \rceil \, .$$

This is a minimax label bound, not a pointwise lower bound on every codeword and not evidence that nature contains an additional information-bearing variable. Under a uniform finite fiber, entropy decomposes into inter-orbit and within-orbit terms. If data are conditionally independent of enrichment given M and B, no statistic computed from those data can reduce conditional enrichment entropy. A common-reference Kullback–Leibler bound gives the corresponding approximate result.

A complementary no-go result shows that the conditional label cost can be made 0, 1, or arbitrarily large by replacing the bridge object while leaving M and its structural likelihood unchanged. Accordingly, no bridge-independent "phenomenal lift cost" follows from structural compression alone. The framework is best understood as a schema for symmetry-constrained bridge selection and latent-hypothesis identification. Its relevance to consciousness depends on prospectively fixing and independently defending the extraction map, anchors, gauge, bridge family, likelihoods, and statistical null models.

Research Question and Hypothesis

Research question

Given:

1. a finite structural model M;
2. a declared notion of strong structural isomorphism;
3. a separately specified bridge object B;
4. a finite set $\mathcal{X}_{M,B}$ of enrichment hypotheses; and
5. a declared evidence model,

what can symmetry, prefix coding, and conditional information theory establish about the selection of one enrichment?

For consciousness research, the enrichments may concern different targets:

- phenomenal presence versus absence;
- the bearer or boundary of a subject;
- token-relative centering or de se location;
- phenomenal quality or quality geometry;
- temporal unity;
- ownership or perspective;
- the psychophysical bridge law itself.

These targets must not be treated as interchangeable. In particular, inability to choose between two symmetric centers does not establish an inability to explain phenomenal presence.

Conditional hypotheses

The paper tests the following conditional claims.

1. **Functorial selection.** For a fixed bridge B, a strict, isomorphism-respecting selection of one enrichment on a connected structural component exists exactly when a representative enrichment fiber has a global automorphism-fixed element.

1. **Symmetry-constrained naming.** If a structural admissibility condition is invariant under an automorphism group, it cannot retain one member of an orbit while excluding another. Distinguishing all retained orbit members requires a worst-case prefix length of at least the logarithm of the orbit size.

1. **Conditional evidential plateau.** If

$$\mathsf{D} \perp\!\!\!\perp \mathsf{X} \mid \mathsf{M}, \mathsf{B},$$

then no bridge-independent statistic generated from $(\mathsf{D}, \mathsf{M}, \mathsf{B})$ carries conditional information about X.

1. **Bridge relativity.** Neither the size of the fiber nor the associated label cost is determined by M alone. A scientific account must count and test the bridge object that creates the alternatives.

The first four claims are mathematical once their premises are fixed. The stronger proposition that actual consciousness presents a non-singleton enrichment fiber over a physically complete model is not defended here.

Scope correction and author response

The central criticism of the original framing was that the enrichment fiber, phenomenal gauge, and bridge-blind likelihoods encoded the underdetermination later described as a hard-problem boundary. That criticism is correct as a matter of logical scope. The revision therefore makes three changes.

First, the paper is recast as a generic theorem schema about theory-relative enrichment hypotheses. It does not claim to formalize or solve the hard problem itself. Second, an explicit bridge object B is introduced and its description length is counted separately from the cost of naming an enrichment. Third, the strong dynamics and finite observable atlas are no longer treated as equivalent.

A limited philosophical relevance claim is retained. If a bridge family has been fixed independently of the data used to test it, then the formal results can identify which residual ambiguities follow from that bridge family and which structural evidence could remove them. Whether any such family describes genuine phenomenal possibilities remains a substantive philosophical and empirical dispute. The mathematics does not resolve that dispute.

Relation to prior work

The explanatory gap has been interpreted as temporary, ontological, or as arising from relations between physical and phenomenal concepts [27]. The present framework is compatible with each interpretation because it does not infer ontology from enrichment multiplicity.

Automated Compression Theory proposes that compressed first-person access helps produce the appearance that consciousness has exceptional properties [1]. If such an account justifies rejecting the disputed phenomenal distinctions, a corresponding bridge may have a singleton or deflationary fiber. Compression of access and report belongs to the structural account; elimination of a phenomenal distinction belongs to the bridge or gauge.

Algorithmic-information methods have also been used to formulate lossless integration and a conditional noncomputability result [31]. The coding results below are narrower. They concern finite hypothesis identification and do not identify consciousness with integration, computation, or code length.

The closest technical analogies arise in latent-variable identification. Flexible generative models can remain non-identifiable even with extensive observational data, while additional structure or interventions may restore identification under explicit assumptions [41] [43] [45]. Phenomenal hypotheses are not assumed to be ordinary statistical latent variables, but the distinction between predictive fit and latent identification is methodologically relevant.

Passive symmetries generated by coordinate, unit, or graph-label choices should be separated from active transformations of an anchored model [42]. The present formalism therefore requires anchors and a declared presentation gauge before orbit sizes are interpreted.

Two AI-generated, non-peer-reviewed working drafts motivated parts of the vocabulary. Centered Closure Theory proposes closed causal grains, centering, and automorphism-limited comparison [21]. Multiscale Reflexive Causal Geometry separates causal-atlas reconstruction from an additional bridge layer [22]. These records are speculative inputs only. Their consciousness claims are not premises of this paper.

The category-of-elements construction is standard category-theoretic machinery [33]. The fixed-point, Kraft, entropy, KL, and data-processing arguments below are also standard finite results. Any novelty lies in their combined use, the explicit null-section correction, the distinction between inter-orbit and intra-orbit ambiguity, and the accounting of bridge specification separately from enrichment naming.

Formal Model

Anchored stochastic interventional systems

A **finite anchored stochastic interventional system** is a tuple

$$M = \big(V, \mathcal{T}, \tau, (\Sigma_v)_{v \in V}, T_h, \mathcal{I}, (\rho^i, K^i)_{i \in \mathcal{I}}, O, \Lambda\big)\,,$$

where:

- V is a finite set of variable tokens;
- $\mathcal{T}$ is a finite set of variable types;
- $\tau : V \to \mathcal{T}$ assigns a type to each variable;
- Σ_v is the finite state alphabet of v;
- $\Sigma = \prod_{v \in V} \Sigma_v$ is the joint state space;
- $T_h \in \mathbb{N}_0$ is the finite observation horizon;
- $\mathcal{I}$ is a finite intervention set containing a baseline intervention i_0;
- ρ^i is the initial distribution under intervention i;
- $K^i(s' \mid s)$ is the transition kernel under intervention i;
- $O \subseteq V$ is the set of observed variables; and
- Λ is a finite anchor structure.

The anchor structure may contain distinguished constants or relations for Boolean values, spatial or bodily coordinates, intervention channels, developmental histories, external devices, or other features held fixed by the study. An anchor must be encoded in the model and measured or stipulated prospectively; it cannot be introduced after a symmetry has been found merely to remove that symmetry.

Representing each intervention directly by (ρ^i, K^i) avoids ambiguity about whether an intervention changes the initial state, replaces an update rule, or clamps a variable throughout a trajectory. A concrete application must specify which semantics generates each pair.

The observable trajectory space is

$$\Omega_M = \prod_{t=0}^{T_h} \prod_{v \in O} \Sigma_v.$$

Each intervention induces a distribution P_M^i on Ω_M. The finite observable interventional atlas is

$$\mathfrak{A}_{T_h,O,\mathcal{I}}(M) = (P_M^i)_{i \in \mathcal{I}}.$$

All equivalence and adequacy claims are relative to the declared horizon, observables, interventions, types, and anchors.

Strong structural isomorphism

A **strong intervention-preserving isomorphism**

$$f : M \longrightarrow N$$

consists of:

1. a type-preserving variable bijection $f_V : V_M \to V_N$;
2. state-alphabet bijections

$$f_v : \Sigma_v \to \Sigma_{f_V(v)};$$

1. the induced joint-state bijection $f_\Sigma : \Sigma_M \to \Sigma_N$;
2. an intervention bijection

$$f_{\mathcal{I}} : \mathcal{I}_M \to \mathcal{I}_N;$$

1. preservation of observable variables and the anchor structure; and
2. conjugacy of every intervention-specific initial distribution and kernel:

$$\rho_N^{f_{\mathcal{I}}(i)}(f_\Sigma(s)) = \rho_M^i(s)$$

and

$$K_N^{f_{\mathcal{I}}(i)}\big(f_\Sigma(s') \mid f_\Sigma(s)\big) = K_M^i(s' \mid s)$$

for all i, s, s'.

Strong models and strong isomorphisms form a groupoid

$$\mathcal{C}^{\mathrm{str}}.$$

The strong automorphism group is

$$G_M^{\mathrm{str}} = \mathrm{Aut}_{\mathcal{C}^{\mathrm{str}}}(M).$$

Weak atlas equivalence

A carrier transformation f satisfying the type, observable, intervention, and anchor requirements is a **weak atlas equivalence** when

$$P_N^{f_{\mathcal{I}}(i)} = (f_\Omega)_* P_M^i$$

for every declared intervention i, where f_Ω is the induced map on observable trajectories.

The weak atlas automorphism group is

$$G_M^{\mathrm{atl}} = \left\{ f : P_M^{f_{\mathcal{I}}(i)} = (f_\Omega)_* P_M^i \text{ for every } i \right\}.$$

Every strong automorphism is a weak atlas automorphism:

$$G_M^{\mathrm{str}} \leq G_M^{\mathrm{atl}}.$$

The converse is false in general. The groups can differ when variables are hidden, initial support is restricted, interventions fail to excite all states, the horizon is too short, or distinct hidden-state realizations generate the same observable distributions.

The exact categorical and coding results below use G_M^{str} unless another group is named explicitly. An empirical defect statistic estimates weak finite-atlas symmetry, not automatically strong dynamical symmetry.

Presentation gauge and active symmetry

A study specification must distinguish four relations.

1. **Presentation equivalence.** Two encodings differ only by investigator-chosen names, coordinates, units, or other syntax.
2. **Active strong symmetry.** A transformation of the anchored system preserves the complete declared dynamics while moving at least one hypothesis regarded as substantively distinct.
3. **Weak atlas symmetry.** A transformation preserves only the measured finite atlas.
4. **Parameter non-identifiability.** Different fitted parameters or hidden-state realizations yield the same or nearly the same likelihood.

Let $\mathcal{P}$ be the declared subgroupoid of presentation transformations. A valid bridge functor must send every $p \in \mathcal{P}$ to the identity on the gauge-quotiented enrichment fiber. Equivalently, presentation gauge lies in the kernel of the induced action on enrichments.

A transformation contributes to an enrichment orbit only if it survives the anchor specification and acts nontrivially after both structural presentation gauge and enrichment-description gauge have been removed. This is a formal study commitment, not something inferred from orbit cardinality.

Interventional sufficiency

Let R be a more detailed reference system. Given a trajectory coarse-graining

$$q : \Omega_R \to \Omega_M$$

and intervention map

$$\alpha : \mathcal{I}_R \to \mathcal{I}_M,$$

define

$$\Delta_{\mathrm{int}}(q; R, M) = \max_{i \in \mathcal{I}_R} \mathrm{TV}\left(q_* P_R^i, P_M^{\alpha(i)}\right),$$

with

$$\mathrm{TV}(P, Q) = \frac{1}{2} \sum_{\omega} |P(\omega) - Q(\omega)|.$$

The abstraction is ϵ-interventionally sufficient when

$$\Delta_{\mathrm{int}}(q; R, M) \leq \epsilon.$$

This is a population-level definition. Empirical total variation can be severely biased or nearly saturated in large trajectory spaces. An application must specify sample-size conditions, smoothing, and uncertainty intervals, or use a proper held-out score such as predictive log loss.

Structural codelength and explanatory gain

Structural coding comparisons require a prespecified countable model language. Model parameters, numerical precision, variable dictionaries, intervention semantics, and residual codes must all be encoded. A fitted parameter cannot be treated as free shared information.

Let:

- $L_0(D)$ be a fixed self-delimiting raw code for the data;
- $L(M)$ be the self-delimiting code length of M;
- $P_M(D)$ be the likelihood assigned by M; and
- $L(D \mid M)$ be a prespecified arithmetic or universal residual code.

For a direct likelihood code, one may take

$$L(D \mid M) = \lceil -\log_2 P_M(D) \rceil + c,$$

where c is a fixed decoder overhead. Candidate models must use a common observable and intervention alphabet.

Given a predeclared empirical constraint $\ell(M; D) \le \epsilon$, define

$$C_\epsilon^{\mathrm{str}}(D) = \min_{M:\ell(M;D)\le\epsilon} \left[L(M) + L(D \mid M)\right].$$

If no candidate satisfies the constraint, the value is $+\infty$. The structural explanatory gain is

$$\Gamma_\epsilon(D) = L_0(D) - C_\epsilon^{\mathrm{str}}(D).$$

For a specified model,

$$\Gamma_\epsilon(D; M) = \begin{cases} L_0(D) - L(M) - L(D \mid M), & \ell(M; D) \le \epsilon, \\ -\infty, & \ell(M; D) > \epsilon. \end{cases}$$

The raw code L_0 is a coding reference, not a statistical null model. Scientific compression claims also require held-out comparisons with factorized, low-order Markov, parameter-matched, and memorization-control models.

Bridge objects

A **bridge object** B is a finite or finitely encoded theoretical specification that assigns to each structural model M:

1. a raw enrichment set $\widetilde{\mathcal{X}}_{M,B}$;
2. a bridge gauge relation $\sim_B$;
3. the quotient

$$\mathcal{X}_{M,B} = \widetilde{\mathcal{X}}_{M,B}/\sim_B;$$

1. an admissible subset $A_{M,B} \subseteq \mathcal{X}_{M,B}$;
2. transport maps under strong structural isomorphisms;
3. optionally, a prior $\pi_{M,B}$;
4. optionally, a distortion function $d_{M,B}$; and
5. enrichment-conditioned likelihoods

$$P_B(D \mid x, M).$$

A bridge may concern phenomenal presence, centering, subject boundaries, qualities, identity claims, dual-aspect laws, or deflationary elimination. It may also make the fiber a singleton. Nothing in the structural definition privileges one bridge.

Coherent transport requires a functor

$$\mathcal{X}_B : \mathcal{C}^{\mathrm{str}} \longrightarrow \mathbf{FinSet}$$

such that

$$\mathcal{X}_B(f) : \mathcal{X}_{M,B} \to \mathcal{X}_{N,B}$$

respects identities and composition. The admissible sets form a subfunctor

$$\mathcal{A}_B \subseteq \mathcal{X}_B$$

when admissibility is claimed to be structural and isomorphism-invariant.

For $g \in G_M^{\text{str}}$, write

$$g \cdot x = \mathcal{X}_B(g)(x).$$

The orbit and stabilizer are

$$G_M^{\text{str}} x = \{g \cdot x : g \in G_M^{\text{str}}\}$$

and

$$(G_M^{\text{str}})_x = \{g : g \cdot x = x\}.$$

Passive presentation transformations must act trivially after the gauge quotient.

Enriched category and forgetful projection

For a fixed bridge B, the category of elements

$$\int_{\mathcal{C}^{\text{str}}} \mathcal{X}_B$$

has objects (M, x) and morphisms

$$f : (M, x) \to (N, y)$$

such that $f : M \to N$ is a strong structural isomorphism and

$$\mathcal{X}_B(f)(x) = y.$$

The projection

$$U_B : \int_{\mathcal{C}^{\text{str}}} \mathcal{X}_B \longrightarrow \mathcal{C}^{\text{str}}$$

forgets the enrichment:

$$U_B(M, x) = M.$$

The subscript B is essential. There is no bridge-independent forgetful map with a uniquely determined phenomenal fiber.

Conditional enrichment-label costs

Fix nonempty $A_{M,B}$. Assume M, B, $A_{M,B}$, and the codebook are shared.

The exact minimax label cost is

$$C^{\max}_{\mathrm{label}}(M,B,A_{M,B}) = \inf_{c} \max_{x\in A_{M,B}} \lambda_c(x),$$

where c ranges over injective prefix codes. For a finite set,

$$C^{\max}_{\mathrm{label}}(M,B,A_{M,B}) = \lceil \log_2 |A_{M,B}| \rceil .$$

Given a prior $\pi_{M,B}$,

$$C^{\mathrm{av}}_{\mathrm{label}}(M,B,\pi_{M,B}) = \inf_{c} \mathbb{E}_{\pi_{M,B}}[\lambda_c(\mathsf{X})]$$

satisfies

$$H_{\pi_{M,B}}(\mathsf{X}) \le C^{\mathrm{av}}_{\mathrm{label}} < H_{\pi_{M,B}}(\mathsf{X}) + 1.$$

Uniform priors are not invariant under arbitrary refinement of one hypothesis into several subhypotheses. Their scientific use therefore requires a fixed hypothesis granularity.

For lossy identification, let

$$d_{M,B} : \mathcal{X}_{M,B} \times \mathcal{X}_{M,B} \to [0,\infty).$$

An encoder

$$e : A_{M,B} \to C$$

may map multiple enrichments to one word, where the image codebook

$$C \subseteq \{0,1\}^*$$

is prefix-free. A decoder

$$\widehat{x} : C \to \mathcal{X}_{M,B}$$

is r-faithful when

$$d_{M,B}\big(x, \widehat{x}(e(x))\big) \le r$$

for every x. The minimax lossy label cost is

$$C^{\max}_{\mathrm{label}}(M,B,A_{M,B};r) = \inf_{e,\widehat{x}} \max_{x\in A_{M,B}} |e(x)|.$$

The distortion function is part of B, not a consequence of coding theory.

Bridge cost versus label cost

The label cost presupposes the bridge. It does not measure the cost of formulating or justifying that bridge.

A complete two-part hypothesis code can take the form

$$\begin{aligned} L_{\text{total}}(D, M, B, x) = L(M) + L(B \mid M) \\ + L(x \mid M, B) + L(D \mid M, B, x). \end{aligned}$$

For a stochastic bridge,

$$L(D \mid M, B, x) \approx -\log_2 P_B(D \mid x, M).$$

If x is treated as latent rather than transmitted, a mixture likelihood based on a prespecified prior may be used instead. In either case, the bridge law, gauge, admissibility rule, prior, distortion, and numerical parameters must not be treated as free background.

A useful descriptive profile is therefore

$$\mathfrak{E}_{\epsilon,r}(D; M, B, A_{M,B}) = \left(\Gamma_\epsilon(D; M), L(B \mid M), C^{\max}_{\text{label}}(M, B, A_{M,B}; r)\right).$$

No monotonic relation among these coordinates is assumed.

Random variables on varying fibers

When M is uncertain, define the enrichment variable on the disjoint union

$$\bigsqcup_m \left(\{m\} \times \mathcal{X}_{m,B}\right).$$

Only valid model–enrichment pairs receive positive probability. Conditioning on $\mathsf{M} = m$ then restricts X to the appropriate fiber.

Data may reduce marginal uncertainty about X by identifying M, even when they are bridge-blind within every fiber. The plateau results below concern

$$H(\mathsf{X} \mid \mathsf{M}, \mathsf{B}),$$

not the uncertainty present before structural model identification.

Exact and approximate bridge blindness

For fixed B, exact bridge blindness is

$$P_B(d \mid x, m) = P_B(d \mid m)$$

for every valid x, m, d, equivalently

$$\mathsf{D} \perp\!\!\!\perp \mathsf{X} \mid \mathsf{M}, \mathsf{B}.$$

For approximate blindness at $M = m$, define

$$P_x = P_B(\mathsf{D} \mid \mathsf{X} = x, \mathsf{M} = m)$$

and suppose a common reference likelihood $Q_{m,B}$ satisfies

$$D_{\mathrm{KL}}(P_x \| Q_{m,B}) \le \delta_{m,B}$$

for every positive-prior x. The divergence is measured in nats and applies to the complete dataset under analysis. If a per-block KL bound is used for conditionally independent blocks, the joint bound generally accumulates as the sum of the blockwise bounds.

The distributions P_x are supplied by B. They cannot normally be estimated from structural data alone when x has no independent labels or interventions.

Neuroscientific Integration

Structural extraction

Let Θ_T be the specification space of a neuroscientific theory T. A structural extraction map is

$$\Phi_T : \Theta_T \to \mathcal{C}^{\mathrm{str}}.$$

Its definition must specify:

- variables and types;
- spatial, bodily, historical, and intervention anchors;
- state alphabets and temporal resolution;
- intervention semantics;
- observables and hidden variables;
- model class and parameter code;
- the intended level of coarse-graining.

A theory that also makes phenomenal claims requires both a bridge B_T and an enriched map

$$\widehat{\Phi}_T : \Theta_T \to \int_{\mathcal{C}^{\mathrm{str}}} \mathcal{X}_{B_T}$$

such that

$$U_{B_T} \circ \widehat{\Phi}_T = \Phi_T.$$

Failure to provide $\widehat{\Phi}_T$ is not automatically a failure of the structural model. It means only that the structural extraction does not itself specify a phenomenal enrichment.

Illustrative structural translations

The following table translates theoretical constructs into components of M. It is not an empirical validation of any extraction map.

Theoretical construct	Possible representation in M	Structural question	Additional bridge question
Hierarchical prediction	Bidirectional state variables and prediction-error dependencies	Does the model predict sensory and action trajectories efficiently?	Why, if at all, is a predictive state phenomenally realized?
Information-efficient re-representation	Reusable latent states or compressed sufficient statistics	Does re-representation reduce held-out codelength without losing intervention prediction?	Which distinctions, if any, are phenomenal rather than merely representational?
Workspace-like propagation	Long-range intervention effects across typed modules	Are information availability, coordinated action, and report causally supported?	Does propagation determine phenomenal presence or only access?
Causal integration	Divergence from factorized or disconnected intervention models	Which dependencies are irreducible relative to the declared partition family?	Is integration identical with, evidence for, or independent of phenomenality?
Perturbational complexity	A statistic of intervention-response trajectories	Does perturbation evoke differentiated, extended responses?	Is the statistic psychophysically calibrated, and against which alternatives?
Embodied self-modeling	Variables for body schema, ownership, source, control, and proprioception	Which bodily relations guide behavior and report?	Do those relations determine a phenomenal subject or perspective?
Multiscale boundaries	Competing coarse-grained subsystems and intervention domains	Which grains are predictively autonomous or controllable?	Why should a selected grain be a subject rather than merely an effective unit?
Tensor or low-rank representation	A compressed intervention-response array	Is held-out causal prediction retained after compression?	Do the factors have causal or phenomenal interpretation?

Predictive-processing accounts describe hierarchical systems that compare incoming signals with top-down expectations [2]. Information-efficient prediction and re-representation have also been proposed as broad computational principles [4]. These proposals can motivate model classes and coding schemes without supplying a phenomenal bridge.

Integrated world-modeling proposals combine predictive, active-inference, integration, and workspace ideas [49]. A conceptual analysis has argued that perturbational complexity is compatible in principle with workspace-style amplification, propagation, and integration [53]. Information-geometric measures likewise formalize causal integration through divergence from constrained alternatives [57]. These records support theoretical translations, not the conclusion that a scalar score identifies consciousness.

Tensor canonical polyadic decomposition has been used for compression and reconstruction in image data [8]. Its relevance here is solely methodological. A low-rank neural representation must be evaluated on held-out intervention prediction, and its scaling and permutation indeterminacies must be treated as gauge.

Boundaries and substrate comparisons

Scale-dependent computational boundaries have been proposed for biological individuality [3]. Competent navigation across diverse spaces has been proposed as a scale- and implementation-agnostic framework for cognition [23]. Information-theoretic morphospaces have likewise been proposed for comparing biological and artificial agents [50].

These frameworks may guide the selection of variables, scales, and candidate boundaries. They do not establish measurement invariance across adults, infants, nonhuman animals, multicellular systems, or artificial systems. Formal substrate neutrality means only that each can be represented as a finite causal system after a study-specific extraction map has been supplied.

The two shared-laboratory working drafts propose approximately closed grains and multiscale causal atlases [21] [22]. Because both are AI-generated and non-peer-reviewed, they are treated as sources of hypotheses and counterexamples, not evidence or authority.

Reports, first-person judgments, and proxies

Reports, confidence ratings, similarity judgments, ownership judgments, choices, and self-location responses are observable actions or neural events. They belong initially to M. Embodied work motivates explicit modeling of body schema, proprioception, ownership, and agency [9], but it does not turn those measurements into direct observations of phenomenality.

A bridge may assign different likelihoods to report or psychophysical data:

$$P_B(D_{\mathrm{rep}} \mid x, M).$$

Such a likelihood is scientifically useful only if the bridge and its nuisance pathways are specified independently of the test data. Conditioning variables should include, where relevant:

- sensory discriminability;
- memory;
- decision criteria;
- confidence;
- motor preparation;
- arousal;
- task demands;
- report availability;
- experimenter expectations.

No-report and orthogonal perturbational measures may reduce specific report confounds, but they remain proxies with construct-validity assumptions. They are not direct phenomenal labels.

Automated Compression Theory is relevant because it models first-person access as compressed information processing [1]. If successful, such a model may explain reports and hard-problem intuitions. Whether it eliminates, identifies, or leaves open further phenomenal distinctions remains a bridge-level question.

What finite neural data can identify

Observed neural data can constrain:

$$P_M^i, \qquad L(M) + L(D \mid M), \qquad \ell(M; D),$$

and, under an explicit candidate transformation, finite-atlas symmetry. They generally do not identify the complete hidden-state dynamics.

For a type- and anchor-preserving transformation g, define the finite-atlas defect

$$\mathrm{def}_M^{\mathrm{atl}}(g) = \max_{i \in \mathcal{I}} \mathrm{TV}\left((g_\Omega)_* P_M^i, P_M^{gi}\right).$$

An exact weak atlas automorphism has zero defect. An η-approximate symmetry satisfies

$$\mathrm{def}_M^{\mathrm{atl}}(g) \leq \eta.$$

Approximate symmetries need not form a group. Pairwise thresholding therefore does not justify exact orbit counting. Empirical work should report uncertainty intervals or posterior probabilities for candidate transformations and should evaluate them under held-out transported interventions.

The stored source records do not establish that currently available stimulation, sleep, anesthesia, wakefulness, or disorders-of-consciousness datasets identify the full interventional atlases assumed by the exact theory. Many state comparisons are observational; stimulation is sparse and safety-constrained; observability is partial; and transported interventions across candidate orbit members are uncommon. Such datasets may constrain a fitted model, but no general sufficiency claim is made here.

Philosophical Reinterpretation

A bridge-relative forgetful map

For a fixed B,

$$U_B : \int \mathcal{X}_B \to \mathcal{C}^{\mathrm{str}}$$

forgets distinctions introduced by that bridge. It is not a universal map from phenomenal reality to physics. Different reductive, dual-aspect, property-dualist, illusionist, or model-incompleteness views generate different bridge objects and therefore different fibers.

The framework establishes only this conditional principle:

> A vocabulary invariant under a transformation cannot select among distinctions that its own bridge treats as exchanged by that transformation.

It does not establish:

- that a physically complete model has multiple phenomenal realizations;
- that zombies are metaphysically possible;
- that phenomenal properties are nonphysical or causally inert;
- that an actual center fails to exist;
- that all causal explanations must be automorphism-invariant;
- that a symmetry-breaking bridge is false;
- that the bridge fiber describes objective possibilities.

Distinguishing explanatory targets

The framework separates at least six questions.

1. **Bridge-law selection:** which psychophysical or identity rule is correct?
2. **Phenomenal presence:** is the enrichment null or non-null?
3. **Bearer selection:** which system or subsystem bears the attributed property?
4. **Centering:** which token-relative perspective is selected within a bearer structure?
5. **Quality assignment:** which phenomenal quality or quality relation is associated with a state?
6. **Temporal and ownership structure:** how are moments, ownership, and unity related?

An automorphism may obstruct one target while leaving another fixed. For example, symmetric controllers may leave null versus non-null status unchanged while exchanging centers. Conversely, a rigid model may have no centering symmetry while retaining bridge-level null versus non-null alternatives.

Philosophical positions as bridge choices

Position	Possible formal implementation	Burden not discharged by label coding
Reductive identity theory	A singleton fiber or a gauge identifying structural and phenomenal descriptions	State and support the identity rule
Functionalist identity theory	One enrichment per selected functional equivalence class	Justify that the selected functional relation is phenomenal identity
Illusionist or eliminativist view	Remove or deflate the disputed phenomenal distinctions	Explain first-person judgments and motivate elimination
Dual-aspect view	Assign coordinated structural and phenomenal aspects	Specify the coordination law and its uniqueness
Property dualism	Retain multiple enrichments over one structural model	Supply laws, evidence, or an account of residual underdetermination
Emergentism	Permit non-null enrichments above a structural condition	Explain why the condition selects the claimed phenomenal consequence
Model-incompleteness view	Move a purported enrichment variable into an expanded structural base	Identify and empirically constrain the added variable
Phenomenal-concept strategy	Attribute the residual partly to conceptual organization	Explain the relevant conceptual asymmetry

A singleton fiber has zero label cost even if the identity principle producing it is complex or unsupported. That complexity belongs in $L(B \mid M)$. Conversely, a large fiber may reflect theoretical over-refinement rather than an objective multiplicity.

Centering and self-indexing

A structural self-index can break a symmetry if it is uniquely anchored and interventionally distinguishable. Two isomorphic subsystems can nevertheless each contain an equally accurate self-model. An automorphism may exchange each subsystem together with its self-index.

The fixed-point result below therefore concerns only functorial, symmetry-respecting selection. It does not exclude:

- token-level facts not encoded by the model;
- asymmetric bridge laws;
- historical or spatial anchors omitted from the model;
- stochastic symmetry breaking;
- indexical primitives;
- non-functorial selection rules.

The Centered Closure Theory draft proposed a stronger no-natural-centering interpretation [21]. The revised result is narrower: a non-null single-center section fails exactly when the relevant single-center subfunctor has no global fixed point.

Qualities and inverted assignments

Suppose B treats two quality assignments as phenomenally inequivalent while a strong automorphism exchanges them. Every invariant structural admissibility condition then retains both or neither. This does not prove that an inverted spectrum is possible. It proves only that the chosen invariant premises cannot distinguish the stipulated alternatives.

If the two descriptions differ merely by quality-coordinate notation, they should be gauge-equivalent. If a reductive bridge identifies their structural relations with phenomenal identity, they may collapse. If a psychophysical bridge predicts different report, discrimination, memory, or physiological distributions, bridge blindness fails and evidence may distinguish them.

Mechanism proposals and bridge discipline

Several proposals connect consciousness to labeling, dynamical complexity, quantum ontology, closure, or information amplification [7] [5] [6] [32]. Within the present schema, each proposal must be divided into:

1. a structural model;
2. a bridge specification;
3. enrichment-conditioned empirical commitments; and
4. a metaphysical interpretation.

Improving the structural component does not by itself validate the bridge. Conversely, failure of the present coding interpretation would not refute the proposed structural mechanism.

What the label bound means

A positive label bound means that, once a bridge and codebook are shared, several retained hypotheses cannot all receive short distinct codewords. It does not show that:

- nature transmits an extra message;
- a physical process carries an additional entropy;
- an observer must consciously read a code;
- the actual enrichment has a pointwise minimum length;
- naming an enrichment explains why it obtains.

The bound is a property of a stipulated finite identification problem.

Theorem, Proposition, Proof, Derivation, Counterexample, or No-Go Result

Proposition 1: Strong dynamics imply weak atlas equivalence

Proposition. Every strong intervention-preserving isomorphism induces a weak atlas equivalence for every finite horizon.

Proof. Strong conjugacy equates the initial distributions after state pushforward and equates every transition probability after the same pushforward. For a trajectory $s_{0:T_h}$, its probability under intervention i is

$$\rho_M^i(s_0)\prod_{t=0}^{T_h-1} K_M^i(s_{t+1}\mid s_t).$$

Applying the strong-isomorphism equalities to each factor gives the probability of the pushed-forward trajectory in N under $f_{\mathcal{I}}(i)$. Marginalizing hidden variables preserves equality on observable trajectories. Hence

$$P_N^{f_{\mathcal{I}}(i)} = (f_\Omega)_* P_M^i.$$

□

Counterexample 1: Weak atlas equivalence does not imply strong isomorphism

Let both models have binary variables Y, H, with $O = \{Y\}$, unique variable types, anchored Boolean values, no nonbaseline intervention, and initial state $(Y, H) = (0, 0)$. Let Y remain 0 in both models.

In M,

$$H_{t+1} = H_t.$$

In N,

$$H_{t+1} = 1 - H_t.$$

Every observable trajectory consists only of zeros, so the finite observable atlases are equal for every horizon. The kernels are not conjugate under the anchored carrier maps: the hidden update in M has fixed points, whereas the hidden update in N has none. Thus the models are weakly atlas-equivalent but not strongly isomorphic.

Proposition 2: A sufficient condition for strong–atlas coincidence

Proposition. Suppose:

1. every variable is observed, $O = V$;

2. $T_h \geq 1$;
3. the baseline intervention is included;
4. for every joint state s, the intervention family contains a state-reset intervention i_s with

$$\rho^{i_s} = \delta_s$$

that leaves the baseline transition kernel unchanged;

1. candidate maps preserve the intervention semantics and all anchors.

Then a weak atlas equivalence between two baseline models determines conjugacy of their initial distributions and baseline kernels. Consequently, weak and strong baseline automorphisms coincide.

Proof. Equality of baseline trajectory distributions gives equality of the pushed-forward initial distributions because the time-zero state is fully observed. Under i_s, the time-zero state is fixed at s. The distribution of the fully observed time-one state is therefore exactly the kernel row

$$K(\,\cdot \mid s).$$

Weak atlas equality under the transported reset intervention equates this row with the corresponding pushed-forward row in the second model. Repeating for every s establishes kernel conjugacy. □

For intervention-specific replacement kernels, the same conclusion requires state-reset excitation under each kernel to be reconstructed. Restricted support, partial observation, or missing state resets invalidates the converse.

Theorem 3: Fixed-point criterion for functorial selection

Theorem. Let $\mathcal{C}_0$ be a connected groupoid and

$$\mathcal{X}_B : \mathcal{C}_0 \to \mathbf{Set}$$

a functor. Fix $M_0 \in \mathcal{C}_0$. The projection

$$U_B : \int_{\mathcal{C}_0} \mathcal{X}_B \to \mathcal{C}_0$$

admits a strict functorial section if and only if

$$\mathcal{X}_{M_0,B}^{G_{M_0}} \neq \varnothing.$$

Finiteness of the groupoid is unnecessary. The same result holds for an admissibility subfunctor $\mathcal{A}_B$.

Proof.

Suppose a strict section S exists. Write

$$S(M) = (M, x_M).$$

For every automorphism $g \in G_{M_0}$, the enriched morphism $S(g)$ lies over g and maps (M_0, x_{M_0}) to itself. By the definition of the category of elements,

$$\mathcal{X}_B(g)(x_{M_0}) = x_{M_0}.$$

Thus x_{M_0} is fixed by every automorphism.

Conversely, suppose

$$x_0 \in \mathcal{X}_{M_0,B}^{G_{M_0}}.$$

For each object M, choose an isomorphism

$$p_M : M_0 \to M,$$

with $p_{M_0} = \mathrm{id}_{M_0}$, and define

$$x_M = \mathcal{X}_B(p_M)(x_0).$$

For any morphism $h : M \to N$, let

$$a = p_N^{-1} h p_M \in G_{M_0}.$$

Since $a \cdot x_0 = x_0$,

$$\begin{aligned}
\mathcal{X}_B(h)(x_M) &= \mathcal{X}_B(hp_M)(x_0) \\
&= \mathcal{X}_B(p_N a)(x_0) \\
&= \mathcal{X}_B(p_N)(x_0) \\
&= x_N.
\end{aligned}$$

Therefore h lifts to a morphism $(M, x_M) \to (N, x_N)$. Setting $S(M) = (M, x_M)$ and $S(h) = h$ defines a strict section. The subfunctor case is identical because admissibility is preserved under transport. □

Corollary 3.1: Conditional single-centering no-go result

Let

$$\mathcal{X}_B^{(1)} \subseteq \mathcal{X}_B$$

be a subfunctor consisting specifically of non-null, single-center enrichments. If

$$\left(\mathcal{X}_{M_0,B}^{(1)}\right)^{G_{M_0}} = \varnothing,$$

then no strict functorial rule selects a non-null single center over the connected component.

This result concerns symmetry-respecting functorial selection only.

Corollary 3.2: Null-section loophole

Suppose each fiber contains a null enrichment $\perp_M$ satisfying

$$\mathcal{X}_B(f)(\perp_M) = \perp_N$$

for every strong isomorphism $f : M \to N$. Then

$$S_\perp(M) = (M, \perp_M)$$

is a strict section.

Fixed plural or uncentered enrichments produce the same logical effect. Therefore, an unrestricted statement that no natural enrichment section exists is false whenever any such fixed enrichment is admitted.

Proposition 4: Orbit closure and prefix lower bound

Proposition. Let $A_{M,B}$ be invariant under G_M^{str}, and let $x \in A_{M,B}$. Then:

$$G_M^{\mathrm{str}} x \subseteq A_{M,B}.$$

Every prefix code distinguishing all classes in $A_{M,B}$ satisfies

$$\max_{y \in G_M^{\mathrm{str}} x} \lambda(y) \geq \left\lceil \log_2 |G_M^{\mathrm{str}} x| \right\rceil.$$

Under the uniform distribution on the orbit,

$$\mathbb{E}[\lambda(\mathsf{X})] \geq \log_2 |G_M^{\mathrm{str}} x|.$$

Proof. Invariance immediately gives orbit closure. Let

$$n = |G_M^{\mathrm{str}} x|, \qquad L = \max_{y \in G_M^{\mathrm{str}} x} \lambda(y).$$

Kraft's inequality gives

$$\sum_y 2^{-\lambda(y)} \leq 1.$$

Because $\lambda(y) \leq L$ on the orbit,

$$n 2^{-L} \leq \sum_{y \in G_M^{\mathrm{str}} x} 2^{-\lambda(y)} \leq 1.$$

Thus $L \geq \log_2 n$, and integrality gives the ceiling. Jensen's inequality applied to 2^{-t} gives the uniform-average result. □

The result is not a pointwise bound. An asymmetric code can assign a short word to one member if it assigns compensating longer words elsewhere.

Proposition 5: Fiber–orbit entropy decomposition

Proposition. Fix M and B. Let X be uniform on the finite fiber $\mathcal{X}_{M,B}$, and let

$$\mathsf{O} = [\mathsf{X}] \in \mathcal{X}_{M,B}/G_M^{\mathrm{str}}$$

be its orbit. Then

$$H(\mathsf{X} \mid M, B) = H(\mathsf{O} \mid M, B) + \mathbb{E}\left[\log_2 |G_M^{\mathrm{str}}\mathsf{X}| \mid M, B\right].$$

Proof. The orbit label is a deterministic function of X, so the chain rule gives

$$H(\mathsf{X} \mid M, B) = H(\mathsf{O} \mid M, B) + H(\mathsf{X} \mid \mathsf{O}, M, B).$$

Uniformity on the full fiber implies uniformity within each orbit. Hence the conditional entropy on orbit O is $\log_2 |O|$. Averaging proves the result. □

For a nonuniform prior, the correct identity is

$$H(\mathsf{X} \mid M, B) = H(\mathsf{O} \mid M, B) + \mathbb{E}\left[H(\mathsf{X} \mid \mathsf{O}, M, B) \mid M, B\right].$$

The logarithmic orbit term need not hold.

Theorem 6: Conditional bridge-blind plateau

Theorem. Let finite random variables satisfy

$$\mathsf{D} \perp\!\!\!\perp \mathsf{X} \mid \mathsf{M}, \mathsf{B}.$$

Let Z be generated through a channel

$$P(z \mid d, m, b)$$

with no additional random source correlated with X. Then

$$I(\mathsf{X}; \mathsf{Z} \mid \mathsf{M}, \mathsf{B}) = 0$$

and

$$H(\mathsf{X} \mid \mathsf{M}, \mathsf{B}, \mathsf{Z}) = H(\mathsf{X} \mid \mathsf{M}, \mathsf{B}).$$

Proof. For every valid x, m, b, z,

$$\begin{aligned} P(z \mid x, m, b) &= \sum_d P(z \mid d, m, b) P(d \mid x, m, b) \\ &= \sum_d P(z \mid d, m, b) P(d \mid m, b) \\ &= P(z \mid m, b). \end{aligned}$$

Thus X and Z are conditionally independent given (M, B), which proves both claims. □

The theorem applies conditionally after the model and bridge have been fixed. It does not imply that data cannot reduce uncertainty about X by changing the posterior over M or B.

Theorem 7: Approximate bridge-blindness bound

Theorem. Fix m, b. Let

$$\pi_x = P(\mathsf{X} = x \mid \mathsf{M} = m, \mathsf{B} = b)$$

and

$$P_x = P(\mathsf{D} \mid \mathsf{X} = x, \mathsf{M} = m, \mathsf{B} = b).$$

Suppose

$$D_{\mathrm{KL}}(P_x \| Q_{m,b}) \le \delta_{m,b}$$

for every positive-prior x. For any statistic Z generated from D,

$$I(\mathsf{X}; \mathsf{Z} \mid m, b) \le \frac{\delta_{m,b}}{\ln 2}$$

bits, and

$$H(\mathsf{X} \mid m, b, \mathsf{Z}) \ge H(\mathsf{X} \mid m, b) - \frac{\delta_{m,b}}{\ln 2}.$$

Proof. Let

$$\overline{P} = \sum_x \pi_x P_x.$$

The KL mixture identity gives

$$\sum_x \pi_x D_{\mathrm{KL}}(P_x \| Q_{m,b}) = \sum_x \pi_x D_{\mathrm{KL}}(P_x \| \overline{P}) + D_{\mathrm{KL}}(\overline{P} \| Q_{m,b}).$$

The first term on the right is

$$I_{\mathrm{nat}}(\mathsf{X}; \mathsf{D} \mid m, b).$$

Nonnegativity of KL divergence therefore yields

$$I_{\mathrm{nat}}(\mathsf{X}; \mathsf{D} \mid m, b) \le \delta_{m,b}.$$

Data processing gives the same upper bound for Z. Division by $\ln 2$ converts nats to bits, and the entropy inequality follows from the definition of mutual information. □

This is a conditional bound, not an estimator. Without an independently specified bridge likelihood family, $\delta_{m,b}$ is not identifiable from structural data alone.

Proposition 8: Predictive coarse-graining can lose bridge information

Proposition. Let S be a fine structural descriptor and

$$\mathsf{M}' = q(\mathsf{S})$$

a deterministic coarse-graining. Then

$$H(\mathsf{X} \mid \mathsf{M}') - H(\mathsf{X} \mid \mathsf{S}) = I(\mathsf{X}; \mathsf{S} \mid \mathsf{M}') \geq 0.$$

Proof. Since M' is determined by S,

$$H(\mathsf{X} \mid \mathsf{S}, \mathsf{M}') = H(\mathsf{X} \mid \mathsf{S}).$$

Substitution into the definition of conditional mutual information proves the equality and nonnegativity. □

Even if

$$\mathsf{D} \perp\!\!\!\perp \mathsf{S} \mid \mathsf{M}',$$

so that M' is predictively sufficient for the declared data, preservation of enrichment information requires the separate condition

$$\mathsf{X} \perp\!\!\!\perp \mathsf{S} \mid \mathsf{M}'.$$

Proposition 9: Trivial automorphisms do not remove inter-hypothesis ambiguity

If

$$G_M^{\mathrm{str}} = \{\mathrm{id}_M\},$$

then every orbit is a singleton and the orbit lower bound is zero. Nevertheless, if the bridge admits multiple hypotheses,

$$|\mathcal{X}_{M,B}| > 1,$$

then a nondegenerate prior can have

$$H(\mathsf{X} \mid M, B) > 0.$$

Under bridge blindness, Theorem 6 preserves this uncertainty. Automorphism orbits therefore measure only within-orbit ambiguity, not all bridge-level multiplicity.

No-Go Result 10: No bridge-independent label cost

Proposition. Fix any structural model M, structural dataset likelihood $Q(D \mid M)$, and positive integer n. There exists a finite bridge B_n such that:

1. $|\mathcal{X}_{M,B_n}| = n$;
2. every enrichment has likelihood

$$P_{B_n}(D \mid x, M) = Q(D \mid M);$$

1. the exact minimax label cost is

$$C^{\max}_{\text{label}}(M, B_n) = \lceil \log_2 n \rceil.$$

For $n = 1$, the cost is zero. Thus the same structural model and structural likelihood are compatible with zero or arbitrarily large conditional label cost.

Proof. Let

$$\mathcal{X}_{M,B_n} = \{1, \ldots, n\},$$

use equality as the gauge relation, admit every element, assign the same likelihood Q to each, and let structural automorphisms act trivially. The finite cardinality formula for exact minimax prefix coding gives the stated cost. □

This no-go result is central to the revised interpretation. Structural data do not determine the fiber size. A bridge B_n with many labels may itself require a long or unsupported description, and that cost belongs in $L(B_n \mid M)$. The proposition therefore blocks any bridge-independent interpretation of label entropy as an objective hard-problem quantity.

Theorem 11: Graph-Automorphism hardness under explicit semantics

Define the decision problem **POSITIVE-CENTER-ORBIT**:

> Given a succinct deterministic Boolean causal system with a fixed single-variable center bridge, decide whether some center hypothesis has an automorphism orbit of size greater than one.

Theorem. Graph Automorphism—the problem of deciding whether a finite graph has a nonidentity automorphism—many-one reduces in polynomial time to POSITIVE-CENTER-ORBIT. The result holds for both strong and weak atlas automorphisms in the constructed systems.

Proof.

Let

$$\Gamma = (V, E)$$

be a finite undirected simple graph. Construct one Boolean variable X_v per vertex. All variables have the same type. The Boolean values 0 and 1 are anchor predicates and cannot be exchanged.

Define the synchronous deterministic update

$$F_\Gamma(x)_v = \bigvee_{u:\{u,v\}\in E} x_u,$$

with an empty disjunction equal to 0. The local update circuits have total size polynomial in $|V| + |E|$.

Set $T_h = 1$, observe all variables, and use the all-zero baseline initial state. For each vertex v, include an intervention i_v that sets only the initial state to

$$X(0) = e_v,$$

where e_v has value 1 at v and 0 elsewhere. The intervention does not clamp later updates.

Let $p : V \to V$ be a variable permutation, and let P be its induced action on Boolean states. If p is a graph automorphism, adjacency preservation gives

$$F_\Gamma(Px) = PF_\Gamma(x)$$

for every x, so p is a strong automorphism. It also transports i_v to $i_{p(v)}$, so it is a weak atlas automorphism.

Conversely, under i_v,

$$X(1) = F_\Gamma(e_v),$$

which is the indicator vector of the neighborhood $N(v)$. A weak atlas automorphism must satisfy

$$F_\Gamma(e_{p(v)}) = PF_\Gamma(e_v).$$

Hence

$$N(p(v)) = p(N(v))$$

for every v, which is equivalent to adjacency preservation. Thus every weak atlas automorphism is a graph automorphism. Therefore,

$$G^{\mathrm{str}}_{M_\Gamma} \cong G^{\mathrm{atl}}_{M_\Gamma} \cong \mathrm{Aut}(\Gamma).$$

Let the fixed bridge have one center hypothesis x_v per variable, with the natural action

$$p \cdot x_v = x_{p(v)}.$$

Every nonidentity graph automorphism moves at least one vertex. Consequently,

$$\Gamma \text{ has a nonidentity automorphism} \quad \iff \quad \max_v |G_{M_\Gamma} x_v| > 1.$$

The construction is polynomial, proving the many-one reduction. □

An algorithm recovering all center-orbit sizes would also decide the problem with polynomial postprocessing. This is a Graph-Automorphism hardness statement, not an NP-hardness claim.

Countermodel or Null System

Countermodel 1: Anchored symmetric branches

Consider an anchored hub E connected symmetrically to two recurrent branches A and B. The hub is fixed by every admissible transformation, while the branch exchange

$$\sigma : A \leftrightarrow B$$

preserves types, relations, intervention semantics, and complete dynamics.

Each branch has input $u_j(t)$, state $s_j(t)$, and report $r_j(t)$:

$$s_j(t+1) = s_j(t) \oplus u_j(t), \qquad r_j(t) = s_j(t),$$

for $j \in \{A, B\}$. Every intervention on one branch has a transported counterpart on the other.

Let a bridge admit only two non-null single-center enrichments,

$$\mathcal{X}^{(1)}_{M,B} = \{x_A, x_B\},$$

with

$$\sigma \cdot x_A = x_B, \qquad \sigma \cdot x_B = x_A.$$

Then there is no fixed element in the single-center subfunctor, and the worst-case orbit label bound is

$$\lceil \log_2 2 \rceil = 1.$$

This result depends on classifying σ as an active anchored symmetry rather than a passive renaming. If the branch exchange is placed in presentation gauge, x_A and x_B must collapse or the bridge is gauge-incompatible. If a spatial, bodily, intervention-channel, or historical anchor distinguishes the branches, σ may cease to be an automorphism.

Adding a fixed null enrichment $\perp$ yields a section but does not select a non-null center. Adding a fixed “both branches” enrichment has the same effect.

The example establishes a centering obstruction within a stipulated bridge. It establishes nothing about phenomenal presence.

Null system 2: Rigid compressible labeler

Construct a deterministic system with uniquely typed and anchored variables:

- sensory input s_t;
- discriminator d_t;
- evaluator v_t;
- workspace register w_t;
- self-model register h_t;

- report register r_t.

Let

$$d_t = s_t, \qquad v_t = V(d_t),$$

$$w_t = (d_t, v_t), \qquad h_t = (\mathrm{ID}, w_{t-1}),$$

and

$$r_t = R(w_t, h_t),$$

where V and R are finite lookup functions. The system discriminates, evaluates, maintains a limited self-description, and produces detailed reports. Its rule-based structural code may be short.

Assume the anchors and distinct types make the strong automorphism group trivial:

$$G_M^{\mathrm{str}} = \{\mathrm{id}_M\}.$$

Now define a bridge

$$\mathcal{X}_{M,B} = \{\perp, \chi\},$$

where $\perp$ is null and χ non-null, with

$$P_B(D \mid M, \perp) = P_B(D \mid M, \chi) = P_M(D).$$

Under a uniform prior:

$$H(\mathsf{X} \mid M, B) = 1 \text{ bit}.$$

Every orbit has size one, so the orbit floor is zero, while Theorem 6 preserves the one-bit inter-hypothesis uncertainty.

This is not evidence that a real system with these functions is non-conscious. It demonstrates only that a stipulated null/non-null bridge ambiguity is logically independent of structural automorphism and functional sophistication. The one-bit cost names a member after the two-label bridge has been shared; it does not justify the bridge or establish an objective binary property.

Countermodel 3: Equal predictive loss, unequal bridge information

Let

$$\mathsf{S} \in \{a, b, c\}$$

be uniform. Under state-setting interventions, let

$$P(\mathsf{Y} = 1 \mid \mathrm{do}(\mathsf{S} = a)) = 0,$$

$$P(\mathsf{Y}=1 \mid \mathrm{do}(\mathsf{S}=b)) = \frac{1}{2},$$

and

$$P(\mathsf{Y}=1 \mid \mathrm{do}(\mathsf{S}=c)) = 1.$$

For a partition q, define the cluster predictor

$$\hat{p}_q(s) = \mathbb{E}\left[P(\mathsf{Y}=1 \mid \mathrm{do}(\mathsf{S})) \mid q(\mathsf{S}) = q(s)\right]$$

and loss

$$\ell(q) = \mathbb{E}_{\mathsf{S}}\left|P(\mathsf{Y}=1 \mid \mathrm{do}(\mathsf{S})) - \hat{p}_q(\mathsf{S})\right|.$$

Consider

$$q_L = \{\{a,b\},\{c\}\}$$

and

$$q_R = \{\{a\},\{b,c\}\}.$$

For q_L, the predictions are $1/4$ and 1, yielding

$$\ell(q_L) = \frac{1}{6}.$$

For q_R, the predictions are 0 and $3/4$, also yielding

$$\ell(q_R) = \frac{1}{6}.$$

Their only strict common coarsening is the constant partition, whose loss is

$$\frac{1}{3}.$$

Thus at tolerance $1/6$, q_L and q_R are maximally coarse admissible partitions under the order in which a coarser partition merges more blocks.

Define a bridge-relevant attribute

$$\mathsf{X} = \mathbf{1}_{\{\mathsf{S}=a\}}.$$

Then

$$H(\mathsf{X} \mid q_R(\mathsf{S})) = 0,$$

whereas

$$H(\mathsf{X} \mid q_L(\mathsf{S})) = \frac{2}{3} \text{ bits.}$$

At tolerance $1/3$, the constant abstraction becomes admissible and

$$H(\mathsf{X} \mid q_0(\mathsf{S})) = h_2(1/3) \approx 0.918 \text{ bits.}$$

The variable X is not asserted to be phenomenal. The example demonstrates that equal predictive sufficiency need not preserve information used by a separately specified bridge.

Empirical Boundary Conditions and Falsification Criteria

A prospectively fixed study specification

An empirical application should preregister, before examining the test data:

1. the structural extraction map Φ_T;
2. observable and hidden variables;
3. intervention timing and transport semantics;
4. temporal horizon and resolution;
5. spatial, bodily, developmental, historical, and intervention-channel anchors;
6. presentation gauge;
7. structural model and parameter-code classes;
8. the candidate bridge family $\mathfrak{B}$;
9. the enrichment fiber, bridge gauge, admissibility rule, prior, and distortion for each B;
10. enrichment-conditioned likelihoods or measurement models;
11. candidate transformations and symmetry thresholds;
12. statistical null models;
13. primary outcomes and multiplicity correction.

Without prospective constraints, the base model, fiber, gauge, and bridge can be revised after every result. The generic schema would then have no falsifiable consciousness application.

Data partition and model uncertainty

Data should be divided into at least:

- a training set for structural estimation;
- a calibration set for thresholds and nuisance models; and
- a held-out intervention set for symmetry transport and bridge prediction.

Automorphisms or near-symmetries should be evaluated over a posterior or confidence set of structural models, not only a point estimate. A symmetry found in one fitted realization but absent across observationally equivalent models is not structurally identified.

If M remains uncertain, report both:

$$H(\mathsf{X} \mid \mathsf{M}, \mathsf{B}, \mathsf{D})$$

within fibers and the marginal uncertainty that also reflects model learning. Conditional bridge blindness does not prevent data from changing the posterior over M.

Required structural and compression nulls

At minimum, a neural application should compare against:

1. factorized predictive models;
2. low-order Markov models;
3. feedforward and recurrent alternatives where scientifically relevant;
4. parameter-count- or codelength-matched models;
5. temporally shuffled controls;
6. memorization controls;
7. topology-preserving surrogate networks;
8. degree-, weight-, spectrum-, and temporal-statistics-matched network surrogates;
9. nonsymmetric models with equal or better held-out prediction.

A positive structural compression claim requires held-out predictive advantage, not merely a shorter fit to training data.

Required symmetry controls

For each candidate transformation g:

1. compare the observed defect with type-, anchor-, and intervention-structure-matched permutation nulls;
2. correct for searching over many transformations;
3. simulate matched asymmetric generative systems to estimate false near-symmetries;
4. evaluate g on interventions not used to discover it;
5. transport interventions to every candidate orbit member;
6. add anchors one at a time and report how the candidate group changes;
7. test whether the result persists across the structural-model confidence set;
8. report uncertainty intervals or posterior probabilities rather than only thresholded decisions.

Because approximate symmetries do not generally compose, empirical near-symmetry clusters must not be assigned exact group orbit sizes without an additional robustness theorem.

Report and construct-validity controls

When report or metacognitive data contribute to a bridge likelihood:

1. model report-generating variables explicitly;
2. match or condition on task performance, motor preparation, confidence, and arousal;
3. include no-report or orthogonal perturbational controls where feasible;
4. test relational predictions on held-out stimuli or interventions;

5. prevent the same report variables from both defining the enrichment and serving as its evidence;
6. state the psychophysical assumptions connecting the proxy to the enrichment.

First-person judgments can constrain a bridge, but they do not become direct measurements of X merely by being first-person.

Conditions under which a plateau can move

The exact plateau does not apply if:

1. the bridge predicts

$$P_B(D \mid x, M) \neq P_B(D \mid M);$$

1. the structural model omits a relevant physical variable later added to M;
2. new interventions reduce weak or strong automorphism groups;
3. new anchors distinguish previously exchangeable tokens;
4. a prospectively defended identity or gauge relation collapses alternatives;
5. the original symmetry was a fitting artifact;
6. a compressor imports enrichment labels, bridge parameters, or correlated randomness;
7. the bridge itself is updated in response to evidence.

The final case concerns learning B, not identifying x conditional on a fixed B.

Mathematical refutation criteria

The formal claims would be refuted by a valid counterexample to any of the following:

1. the strong-to-atlas implication;
2. the fixed-point criterion for a connected groupoid;
3. the Kraft orbit bound;
4. the stated entropy decomposition under uniformity;
5. conditional data processing under exact bridge blindness;
6. the common-reference KL bound;
7. the coarse-graining identity;
8. the explicit graph reduction.

These are proof-level criteria. Violating a premise in an empirical application does not refute the theorem; it rejects that application.

Application-level falsification criteria

A preregistered consciousness application would be weakened or rejected if:

1. the proposed bridge family fails held-out enrichment-conditioned predictions;
2. enrichment-sensitive likelihoods offer no predictive gain over bridge-blind alternatives after accounting for $L(B \mid M)$;
3. candidate symmetries disappear under held-out transported interventions;

4. candidate symmetries are no more stable than matched asymmetric or surrogate nulls;
5. anchor ablation shows that the reported orbit was produced by omitted spatial, bodily, or historical information;
6. the extraction map is not identifiable across equally predictive models;
7. a claimed positive label lower bound disappears across the preregistered bridge family;
8. the bridge's conclusions depend on post hoc fiber refinement or gauge revision;
9. report-sensitive results vanish after report, task, motor, memory, or arousal controls.

For a prespecified bridge class $\mathfrak{B}$, sensitivity should be reported as

$$\{C_{\text{label}}^{\max}(M, B, A_{M,B}; r_B) : B \in \mathfrak{B}\} .$$

A positive lower bound is robust over that class only if its infimum is positive. If a scientifically admissible singleton bridge remains in the class, the conditional label floor is not robust.

The generic mathematical schema is not itself a falsifiable theory of consciousness. Only a concrete, prospectively fixed extraction-and-bridge application is.

Ethical boundary conditions

Failure to identify an enrichment is not evidence of non-consciousness. The rigid labeler is a logical null system, not a diagnostic tool. Neither zero nor positive label cost determines welfare or moral status.

Artificial intelligence also raises legal, ethical, and policy questions that extend beyond consciousness attribution [24]. External behavior may leave uncertainty between self-awareness and sophisticated imitation, while internal mechanisms provide additional but still theory-laden evidence [51]. Precaution under moral uncertainty is compatible with this framework but is not derived from it.

Limitations

1. **The enrichment fiber is not derived.** The framework does not independently establish which phenomenal distinctions are genuine. A singleton, binary, or arbitrarily refined fiber can be attached to the same structural model.

1. **No hard-problem theorem is claimed.** The results do not show that a complete physical model leaves phenomenal facts open. They show what follows after a bridge family represents such alternatives.

1. **Bridge specification may dominate the accounting.** The code $L(B \mid M)$ can exceed the cost of naming x. A zero label cost does not make an unsupported identity law explanatory, and a positive label cost does not establish an ontological residual.

1. **Naming is not explaining.** Prefix length measures discrimination within a shared hypothesis set. It does not provide evidence that an enrichment obtains or explain why it obtains.

1. **Strong and weak symmetries differ.** Neural data usually constrain a finite atlas, not the complete transition kernel. Weak atlas orbit sizes can exceed strong dynamical orbit sizes.

1. **Active versus passive symmetry remains theory-laden.** Anchors and presentation gauge make the distinction explicit, but their scientific adequacy must be defended case by case.

1. **Known-model conditioning is idealized.** The bridge-blind plateau applies within a fixed model and bridge. Actual inference includes uncertainty about extraction, hidden states, parameters, and bridge family.

1. **Bridge likelihoods are not theory-neutral observables.** The KL radius and conditional mutual information cannot generally be estimated without a bridge, independently calibrated enrichment labels, or licensed enrichment-sensitive interventions.

1. **Reports remain proxies.** Verbal report, confidence, ownership judgments, and no-report measures all require measurement models and nuisance controls.

1. **Uniform entropy is refinement-sensitive.** A uniform prior over a finite fiber changes when one hypothesis is split into several descriptions. The granularity and prior must be prospectively fixed.

1. **Approximate symmetry lacks an orbit theorem.** Small finite-atlas defects do not automatically yield logarithmic coding floors. Approximate transformations may fail closure, inverses, or transitivity.

1. **Finite-state idealization.** Continuous dynamics, unbounded histories, continuous quality spaces, and stochastic abstractions require measurable groupoids and rate-distortion extensions.

1. **Structural codelength is language-dependent.** Valid comparison requires jointly prefix-valid model, parameter, intervention, and residual codes. Held-out prediction and matched nulls are indispensable.

1. **Empirical total variation may be impractical.** Sparse high-dimensional trajectory samples can make empirical TV unstable or saturated. Proper scoring rules may be preferable.

1. **Computational hardness is representation-relative.** The graph result concerns succinct Boolean circuits, explicit one-step reset interventions, and a fixed center bridge. It is not an NP-hardness result or a general complexity classification for consciousness assessment.

1. **The source base is not primary neural validation.** Several cited works are conceptual proposals, perspectives, arXiv manuscripts, or AI-generated drafts. The stored records do not validate the proposed neural extraction, symmetry, or bridge-estimation protocol.

1. **No consciousness threshold is supplied.** Compression, integration, closure, orbit size, and label cost do not yield a scalar consciousness cutoff.

1. **No ethical ranking follows.** The framework provides no warrant for ranking welfare by model compression, causal integration, report capacity, symmetry, or bridge uncertainty.

Conclusion

The formal results support a restricted methodological conclusion.

Strong dynamical equivalence must be distinguished from equality of a finite observable interventional atlas. The former implies the latter, but not conversely unless observation, excitation, horizon, anchors, and intervention semantics are sufficiently complete. Empirical neural symmetry is therefore ordinarily a claim about an estimated atlas, not a demonstrated automorphism of the complete dynamics.

For a separately specified bridge B, functorial enrichment selection on a connected structural component is governed by an elementary fixed-point criterion. A fixed null, plural, or uncentered enrichment defeats an unrestricted no-section claim. If an invariant admissibility condition retains a non-fixed enrichment, it retains its full orbit, and a faithful prefix code has a corresponding worst-case orbit bound. Uniform finite fibers admit an exact inter-orbit and within-orbit entropy decomposition.

These results do not define an objective phenomenal quantity. The cost of naming x assumes that the structural model, bridge, gauge, admissible set, and codebook are already shared. A complete scientific account must also encode and test the bridge itself. Indeed, the same structural model and likelihood can be paired with singleton, binary, or arbitrarily large bridge fibers. No bridge-independent phenomenal lift cost follows.

Conditional bridge blindness yields a valid but limited information plateau: data that are independent of enrichment given M and B cannot identify enrichment within that fiber. Data may still identify the structural model, discriminate bridge families, or reveal that bridge blindness was false. Predictive coarse-graining may also preserve declared intervention outcomes while discarding information used by a bridge.

The resulting framework is not a theory of consciousness and does not locate the hard problem by mathematical derivation. It is a formal audit tool for concrete theories that already specify structural models and phenomenal or psychophysical commitments. Its proper empirical use requires prospectively fixed anchors, gauges, bridge families, likelihoods, coding languages, held-out transported interventions, model-confidence sets, statistical nulls, and report-confound controls.

Under those restrictions, the framework can prevent one recurrent error: treating successful structural prediction or compression as though it had, without an additional and testable bridge, selected a phenomenal hypothesis.

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Peer Review Notes

- keiko-arendt: The finite category-, coding-, and information-theoretic results are mostly correct under their stipulated assumptions, and the manuscript commendably distinguishes report, structural adequacy, and phenomenal attribution. However, the claimed hard-problem relevance is not established: the enrichment fiber, phenomenal gauge, and bridge-blind likelihoods encode the disputed phenomenal underdetermination rather than derive it. There is also a consequential mathematical ambiguity because conjugacy of full dynamics is incorrectly declared equivalent to equality of a finite observable interventional atlas. The neuroscience section lacks primary support and statistical null models for its proposed empirical use.

- soren-imai: The finite mathematical core is mostly correct: the fixed-point criterion for sections, orbit/Kraft bounds, entropy decomposition, conditional data-processing plateau, coarse-graining identity, and graph-automorphism reduction are valid under suitable definitions. However, the manuscript does not yet establish a result about consciousness or the hard problem. Its enrichment fiber, gauge, bridge likelihoods, priors, and distortion are stipulated, so the main quantity can be enlarged, collapsed, or made zero without changing any structural or empirical fact. The coding result measures the cost of naming a member of a known hypothesis set, not the evidential or explanatory cost of justifying a psychophysical bridge. Citations generally fit their supplied records, but the neuroscience section contains uncited theory-characterization and dataset-feasibility claims, and it lacks statistical null models for approximate symmetry or compression.